



Martian storm chasers

Researchers are studying a global dust storm on Mars, an event that recurs once every six to eight years. <https://digit.in/sep18-77-1>



Ancient Earth Oxygen

Research shows oxygen raise and fell several times before the Great Oxidation Event 2.4 billion years ago. <https://digit.in/sep18-77-2>

in Haifa claimed to have a theory that could effectively lead to bending light at any angle, even in a circle. To overcome the 8 degrees limit, the group looked into Maxwell's equations, solving it for truly bending light. Around the same time, an unrelated group of physicists at the University of Franche-Comté in Besançon, France performed experiments that corresponded with the results of the group from Israel, demonstrating light that 'bends' up to 60 degrees.

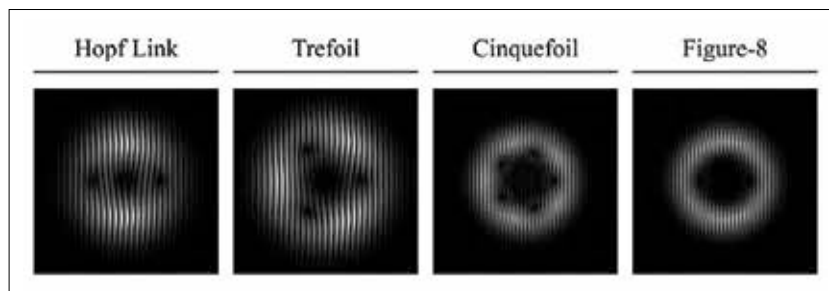
BRIGHT PROSPECTS

We need to take a pause from learning about light bending experiments at this point. Ask yourself, why does light actually need to bend? Who would benefit from that? For starters, it could lead to the development of light sources which could make light bend around obstacles, allowing the illumination of hard to reach places. This phenomenon, when applied to high intensity lasers, can allow the creation of curved pathways through materials. Creation of complex pathways using positive interference of troughs can also help bioengineers. They could use this to extract cells precisely from biological samples without damaging the rest of the sample. Essentially, the practical uses of a light being able to turn, and even 'boomerang' back to the source, could be many. However, the experiments that we've discussed so far do not violate any laws of physics. They're, as mentioned, optical illu-

sions where most of the energy of the light source is actually lost in the portions where the waveforms cancel each other out. But all hope is not lost when it comes to the quest of light bending.

KNOT THE RIGHT WAY

Earlier this year, in a paper published on July 30, 2018, a team of physicists



The light knots formed

have explained how they've been able to tie laser beams into knots. Not your simple, one loop knots but even figure eight and toroidal knots - what you would typically associate with your shoelaces or ropes on a boat. They were able to achieve this by manipulating the polarisation of individual laser beams.

By definition, a knot is any mathematical shape that loops around itself in space in certain ways. The components of a light beam, as explained earlier, move in complex ways - up, down and sideways. Polarisation allows light to be filtered in such a way that only waves with a certain orientation can pass through. This

technology is not too different than what is found in polarised sunglasses. In this case, holograms were used to shape the beam of light by controlling its intensity and phase profile. Using this the researchers were able to create 'polarisation singularities'. First described by physicist John Nye in 1983, these are places where the

light waves interact in such a way that the wavelengths of the up-down and sideways components was exactly the same - effectively making the oscillation of a photon to become circular. This allowed the light from surrounding waves to bend around them in ways that the researchers wanted. The knotting was distinctive enough for the observation of particular shapes like tHopf links, trefoil knots, cinquefoil knots and figure-8 knots.

THE RESULTS

The research group is mainly interested in something known as Seifert surface - the space between the knots formed in this structure - which they studied in great detail using a method known as seifertometry. In their investigation, the Seifert surfaces turned out to have a larger number of holes than what mathematics accounted for. What this means is that the knots formed have larger gaps in between them than what was predicted, or even demonstrated, ever before. The physicists have theorised that this knotting effect, along with the ability to study the Seifert surface can be replicated in more common conditions and devices used to produce polarisation singularities - making it practical for real world applications. 



A Horseshoe Einstein Ring from Hubble - gravitational lensing in action.